FLYBOAT

Flyboat Electronic LLC — North American sales & engineering center

46L Series Metal Oxide Varistors

Epoxy-encapsulated zinc oxide varistors · Datasheet

The 46L series is a line of 46 mm rectangular, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs — the largest disc format in the range. Rated In 30 kA, with Imax 60 kA on the 100–390 V grades and 70 kA on the 430–1,200 V grades (8/20 µs), the standard range covers varistor voltages from 100 V to 1,200 V in 27 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top grades the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Certification status

The 46L series is not UL recognized: the catalog ratings tables for this series print the UL / C-UL / TÜV columns but leave every cell blank. For series not yet UL recognized, we can pursue UL certification for your program.
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Model naming: 46L K 431 — 46 = nominal size (mm); K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead / structure form.

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

46L standard ratings — full table

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
46LK101	100	±10	60	85	30	60	360-3.6	270
46LK121	120	±10	75	100	30	60	400-3.6	330
46LK151	150	±10	95	125	30	60	450-3.0	420
46LK181	180	±10	115	150	30	60	490-2.8	490
46LK201	200	±10	130	170	30	60	520-2.6	560
46LK221	220	±10	140	180	30	60	550-2.6	630
46LK241	240	±10	150	200	30	60	600-2.6	660
46LK271	270	±10	175	225	30	60	680-2.6	760
46LK301	300	±10	195	250	30	60	770-2.6	870
46LK331	330	±10	210	270	30	60	830-2.55	940
46LK361	360	±10	230	300	30	60	910-2.55	1050
46LK391	390	±10	250	320	30	60	1000-2.55	1160
46LK431	430	±10	275	350	30	70	1050-2.45	1320
46LK471	470	±10	300	385	30	70	1100-2.4	1420
46LK511	510	±10	320	410	30	70	1150-2.35	1520
46LK561	560	±10	350	450	30	70	1250-2.35	1650
46LK621	620	±10	385	505	30	70	1450-2.3	1800
46LK681	680	±10	420	560	30	70	1650-2.3	1810
46LK711	710	±10	440	585	30	70	1750-2.3	1830
46LK751	750	±10	460	615	30	70	1800-2.3	1840
46LK781	780	±10	485	640	30	70	1750-2.3	1860
46LK821	820	±10	510	670	30	70	1800-2.3	1960
46LK911	910	±10	550	745	30	70	2100-2.3	21220
46LK951	950	±10	575	765	30	70	2200-2.3	2130
46LK102	1000	±10	625	825	30	70	2400-2.3	2220
46LK112	1100	±10	680	895	30	70	2700-2.3	2435
46LK122	1200	±10	750	1050	30	70	3000-2.3	2535

46L low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
46LK431	430	±10	275	350	30	70	950-2.2	1320

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
46LK471	470	±10	300	385	30	70	1030-2.2	1420
46LK511	510	±10	320	410	30	70	1130-2.2	1520
46LK561	560	±10	350	450	30	70	1240-2.2	1650
46LK621	620	±10	385	505	30	70	1350-2.15	1800
46LK681	680	±10	420	560	30	70	1480-2.15	1810
46LK711	710	±10	440	585	30	70	1540-2.15	1830
46LK751	750	±10	460	615	30	70	1630-2.15	1840
46LK781	780	±10	485	640	30	70	1670-2.15	1860
46LK821	820	±10	510	670	30	70	1850-2.1	1960
46LK911	910	±10	550	745	30	70	1940-2.1	2060
46LK950	950	±10	575	765	30	70	2040-2.1	2130
46LK102	1000	±10	625	825	30	70	2200-2.1	2220
46LK112	1100	±10	680	895	30	70	2380-2.1	2435
46LK122	1200	±10	750	1050	30	70	2620-2.1	2535

46L high-energy ratings (10/350 µs)

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Iimp (kA)	UP (V)	ETM (J)
46LK431	430	±10	275	350	30	8.5	1050	1320
46LK471	470	±10	300	385	30	8.5	1100	1420
46LK511	510	±10	320	410	30	8.5	1150	1520
46LK561	560	±10	350	450	30	8.5	1250	1650
46LK621	620	±10	385	505	30	8.5	1450	1800
46LK681	680	±10	420	560	30	8.5	1600	1810
46LK711	710	±10	440	585	30	8.5	1650	1830

Ratings as published at www.flyboatmov.com/46l-series.html.

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